Lab Experiment. 4

Setting up a LAN

Aim:

To design setup and configure a local area network connecting four computers through a network surface assign static IP address to all devices verify connectivity using the ping command.

Equipment Requirement:-

1. Personal computers - 4 - Ethernet enabled
2. Network switch - 1 - 8/16/24 port
3. Ethernet cables - 6 - straight through RJ-45

IP address configuration Table.

Device	IP address	Subnet mask	connected to
PC_0	10.1.1.1	255.0.0.0	switch port FA01
PC_1	10.1.1.2	255.00.0	switch port fa02
PC_2	10.1.1.3	255.0.0.0	switch port fa013
PC_3	10.1.1.4	255.0.0.0	switch port FO14
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation is CISCO packet Tracer since physical hardware was limited the LAN use is also implemented and verified using cisco packet tracer.

Steps:-

1. Open cisco packet tracer
2. From the device category panel, select find devices and 4 PCs on to workspace
3. Select switches and drag 1 switch
4. Click connections → select copper straight
5. Click each PC and then click the switch and select an available fast ethernet port.
6. Repeat for all four PCs Green link lights confirm successful physical connection.
7. Click a PC → Desktop → IP config
8. Give the static IP address and subnet mask as per the IP table for each PC.
9. Click the switch → CLI tab
10. Find the following commands to assign management IP enable
configure terminal
interface vlan 1
IP address 10.1.1.5 255.0.0.0
shutdown
exit
11. Click PC → Desktop tab → command prompt
12. Use the ping command to test connectivity to the other PCs, eg:- ping 10.1.1.2

13) successful "Reply from".

12. Add server service
13. Assign it an IP address via it's Desktop
14. Config tab
15. Open the server's service tab and HTTP
16. From another PC, open the web Browser and enter the servers IP address to load the page.
18. Switch to simulation mode.
19. Send a ping or web request two devices and use the play to stop through the packet.

Observation:-

All four PC's were successfully connected to the switch on the star topology and assigned an unique static IP address with the same subnet (10.0.0.0/255.0.0.0) the shared server resource was successfully accessed from other computers on the network, confirming that the LAN was functioning correctly.

Result:-

Successfully connected.

Lab Experiment - 5

Aim:-

experiment on packet capture tool: wires
To filter, capture, view, packets in wire
Tool capture 100 packets from the ethern
IEEE 802.3 LAN interface and save it

wire shark:-

wire shark a network analysis tool firmly known as ethernet, captures packets in real time and display them in human-readable format. It includes filters, color coding and other features that let you dig deep into network traffic.

Procedure:-

* select Local area connection wireshark.
* Go to capture option
* select capture automatically after 100 pac
* Then click start capture.
* save the packets.

1) create a filter to display only TCP/UD packets inspect the packets and provide the graph.

Procedure:-

* Select LAN connection to wireshark.
* Go to capture option.
* select stop capture automatically after 100 packets.

* start top package in search bar
* To see flow graph click statics flow graph
* save the packets

2) create a filter it's display only ARP pack and inspect the packets

Procedure:-

* GO to capture option
* select stop capture automatically after 100 packets
* Then click start capture.
* search ARP packets in search bar
* save the packets.

3. create a filter to display only DNS packet and provid the flow graph.

Procedure:-

* Go to capture option
* Select the capture automatically 100 packets.
* Then click start capture.
* search DNS packets in search bar
* To see the flow graph.
* Save the packets.

4. Create a filter to display HTTP packet & inspect the packet:-

procedure:-

* Select local area connection in wireshark.

* to to capture option.
* Then click start capture.
* Search HTTP packets in search bar.
* save the packets.

5. create a files to display only IP/Icmp packets.

Procedure:-

* select local area connection in wireshark.
* Stop capture automatically after 100 packets
* Then click start capture.
* search Icmp/ip packets in search bar. save the packets

6. create a filter to display only Dhcp packets

* select local area connection in wireshark.
* Go to capture option.
* Select stop capture automatically after 100 packets.
* Then click start capture.
* Search Dhcp packets in search bar.
* Share the packets

Result:-

wireshark too using the the capturing and analysing packets are done successfully.